

EXTENDS *Naturals, Sequences*

$$\begin{aligned} \text{Keys} &\triangleq \{ \text{"checking"}, \text{"savings"} \} \\ \text{Values} &\triangleq \{0, 1, 2\} \\ \text{InitVal} &\triangleq 0 \end{aligned}$$
We define the *Workload* as a Sequence (Tuple) of Records.
$$\begin{aligned} \text{Workload} &\triangleq \langle \text{T1: TransactSavings (PSI)} \\ &[\\ &\quad \text{session} \mapsto \text{"client_1"}, \\ &\quad \text{level} \mapsto \text{"PSI"}, \\ &\quad \text{ops} \mapsto \langle [\text{type} \mapsto \text{"R"}, \text{key} \mapsto \text{"checking"}], \\ &\quad \quad [\text{type} \mapsto \text{"R"}, \text{key} \mapsto \text{"savings"}], [\text{type} \mapsto \text{"W"}, \text{key} \mapsto \text{"savings"}, \text{val} \mapsto 1] \rangle \\ &], \text{T2: WriteCheck (SER)} \\ &[\\ &\quad \text{session} \mapsto \text{"client_2"}, \\ &\quad \text{level} \mapsto \text{"SER"}, \\ &\quad \text{ops} \mapsto \langle [\text{type} \mapsto \text{"R"}, \text{key} \mapsto \text{"checking"}], \\ &\quad \quad [\text{type} \mapsto \text{"R"}, \text{key} \mapsto \text{"savings"}], [\text{type} \mapsto \text{"W"}, \text{key} \mapsto \text{"checking"}, \text{val} \mapsto 2] \rangle \\ &] \\ &\rangle \end{aligned}$$

VARIABLES

$$\begin{aligned} &\text{client_dep}, \\ &\text{client_last_commit}, \\ &\text{router_ct}, \\ &\text{shard_ct}, \\ &\text{tx_status}, \\ &\text{tx_snapshot}, \\ &\text{tx_checkset}, \\ &\text{tx_buffer}, \\ &\text{tx_commit_time}, \\ &\text{pc}, \\ &\text{store}, \\ &\text{op_seq} \end{aligned}$$
 $MI \triangleq \text{INSTANCE } \textit{MixedIsolation} \text{ WITH}$

$$\begin{aligned} &\text{Keys} \leftarrow \text{Keys}, \\ &\text{Values} \leftarrow \text{Values}, \\ &\text{InitVal} \leftarrow \text{InitVal}, \\ &\text{Workload} \leftarrow \text{Workload} \end{aligned}$$
 $\text{Spec} \triangleq MI! \text{Spec}$ $\text{OperationalSafety} \triangleq MI! \text{OperationalSafety}$